

Kul'shman's chandhary

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES
Minor Exam

M.CA. (Winter Semester-2024)

Course Name: Computer Architecture
Course Code: CS 106

Maximum Marks: 30
Total Time: 1 hour

Q1. What is the difference between RISC and CISC type of Computer Architecture?.

A computing task is divided into three parts:

1. Part A: 50% of the task can be fully parallelized.
2. Part B: 30% of the task can be parallelized.
3. Part C: 20% of the task is inherently serial and cannot be parallelized.

Calculate the overall speedup when using 8 processors for Part A and 4 processors for Part B. Part C remains serial.

(4+6)

Q2. (a) What are the major performance parameters that determines the execution time of processor?.

b) What is Harvard Architecture?.

- (a) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 1.5. ii) reduce CPI instruction type C to 1.8. Which one is better?.

Type	CPI	Frequency
A	1.3	40%
B	2.3	40%
C	2.1	20%

(5+2+3)

Q3. (a) What are the different types of addressing modes that in 8086 microprocessor?.

- (b) Write an assembly language program to subtract two numbers using 2^{nd} complement. What is the new memory capacity if the RAM memory is scaled upto from 2^{20} byte to 2^{28} byte?.

(4+4+2)